

AIML PROJECT

AI-Based Career Guidance System using NLP and Machine Learning

Abstract

Choosing the right career path is a common challenge for students. This project presents an AI-based system that suggests suitable career options based on a user's interests, skills, and personality described in natural language. The system uses Natural Language Processing (NLP) and Machine Learning techniques to analyze user input and provide multiple career recommendations with matching probabilities. The goal is to create a simple, interactive, and intelligent decision-support system for students.

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Choosing the right career path is a common challenge for students. This project presents an AI-based system that suggests suitable career options based on a user's interests, skills, and personality described in natural language. The system uses Natural Language Processing (NLP) and Machine Learning techniques to analyse user input and provide multiple career recommendations with matching probabilities. The goal is to create a simple, interactive, and intelligent decision-support system for students.

Introduction

With the increasing number of career options available today, students often face confusion in selecting the most appropriate path. Traditional guidance methods may not always be personalized or accessible. Artificial Intelligence (AI) can help solve this problem by analyzing user preferences and providing intelligent recommendations.

This project develops a career guidance system that:

- Accepts user input in natural language
- Processes the input using NLP
- Uses machine learning to predict suitable careers
- Displays top career options with percentage match

Objectives

- To build an AI-based career recommendation system
- To use NLP for understanding user input
- To apply machine learning for prediction
- To provide multiple career suggestions with probabilities
- To create an interactive GUI for ease of use

Scope of the Project

The system can be used by:

- Students for career selection
- Beginners exploring fields
- Educational institutions for guidance

Future improvements may include:

- Larger datasets
- More accurate NLP models
- Web/mobile application

Methodology

1. Data Collection

A dataset (career.csv) is created containing:

- Keywords representing skills/interests
- Corresponding career labels

2. Data Preprocessing

- Text data is cleaned and formatted
- Keywords are structured into meaningful categories

Feature Extraction (NLP)

- TF-IDF (Term Frequency–Inverse Document Frequency) is used
- Converts text into numerical form

4. Model Training

- Algorithm used: **Multinomial Naive Bayes**
- Model is trained on processed data

5. Prediction

- User input is processed
- Model predicts probabilities for careers
- Top 3 careers are displayed

6. GUI Implementation

- Developed using **Tkinter**
- Provides input box and result display
- Styled with colors and emojis for better experience

Technologies Used

- Python
- Pandas
- Scikit-learn
- Tkinter
- TF-IDF Vectorizer
- Naive Bayes Algorithm

System Architecture

1. User Input (Text)
2. Feature Extraction (NLP)
3. Machine Learning Model
4. Prediction
5. Output Display (GUI)

Output

The system displays:

- Top 3 career matches
- Matching percentage
- Reasoning based on input

Example:

Top Career Matches:

Data Scientist → 60%

Software Engineer → 25%

AI Engineer → 15%

Advantages

- Easy to use
- Provides multiple suggestions
- Personalized recommendations
- Interactive GUI
- Covers modern careers

Limitations

- Limited dataset
- Depends on input quality
- Basic NLP (not advanced)

Applications (Unit V Connection)

This project is an application of AI in:

- Education
- Career guidance
- Decision support systems

Conclusion

The project successfully demonstrates how AI can be used to assist students in choosing careers. By combining NLP and machine learning, the system provides meaningful recommendations in a simple and interactive manner. It highlights the potential of AI in solving real-world problems.

Future Scope

- Improve dataset size
- Use deep learning models
- Add voice input
- Develop mobile/web application

References

- Scikit-learn Documentation
- Python Documentation
- AI/ML learning resources

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